

Chapter 1: Coupled Simulation

In this chapter, we couple the locally mass-conserving MFEM and FVM methods to perform numerical simulations of a partially molten mantle. We describe the implementation details and coupling strategy that integrate the MFEM, FVM, and the eutectic phase package, with the remaining uncertainty in porosity ϕ addressed through phase package evaluation. Before tackling the fully coupled problem, we first examine an alternative porosity evolution scenario based on a prescribed melting rate. We then present an example that combines the MFEM, FVM, and eutectic phase package into a fully coupled computation. The discussion begins with pseudo-1D scenarios and subsequently extends to full 2D simulations.

1.1 Coupled Algorithm

Let $\mathcal{U}^n = (C^n, H^n)$ denote the cell-averaged solution to the transport system (??) at time t^n . Similarly, Let $\mathcal{V}^n = (\check{\mathbf{v}}_r^n, \check{\mathbf{v}}_s^n, \check{q}_l^n, \check{q}_s^n)$ represent the solution to the Darcy-Stokes system at the same time step t^n . Finally, let $\mathcal{W}^n = (\phi^n, \phi_1^n, \check{T}^n, c_s^n, c_l^n)$ denote the output of the phase calculation at t^n obtained from the transport result $\mathcal{U}^n$.

For each time step t^n , we first reconstruct dimensionless mass composition and enthalpy from the cell-averaged solution $\mathcal{U}^n$ using ML-WENO reconstruction at each data points. The eutectic phase package then provides the corresponding phase parameters $\mathcal{W}^n$ based on these pointwise values. With $\mathcal{W}^n$ determined, we solve the Darcy-Stokes system at the fixed time t^n to obtain $\mathcal{V}^n$. Finally, the cell-averaged solution $\mathcal{U}^{n+1}$ at the next time step t^{n+1} is then obtained by forward marching of the transport system. The complete procedure is summarized as an coupled solver in algorithm 1.

Algorithm 1 Sequential Coupled Solver

Let U^0 be given as initial data.

- 1: Initialize $\mathcal{U}^0$, compute $\mathcal{W}^0$ with eutectic phase package.
- 2: Compute ϕ^0 in $\mathcal{W}^0$, solve the Darcy-Stokes system (??)–(??) for $\mathcal{V}^0$.
- 3: **for** Time level $n = 1, 2, \dots, n_{\max} - 1$ **do**
- 4: Given $\mathcal{U}^n$, compute $\mathcal{W}^n$ by the eutectic phase package.
- 5: Compute ϕ^n in $\mathcal{W}^n$, solve the Darcy-Stokes system (??)–(??) for $\mathcal{V}^{n+1}$.
- 6: Given $\mathcal{W}^n$ and $\mathcal{V}^{n+1}$, compute dimensionless effective velocity (??) and phase averaged velocity as

$$\check{\mathbf{v}}_{\text{eff}} = \frac{\phi \check{\mathbf{v}}_l + D_i(1 - \phi) \check{\mathbf{v}}_s}{\phi + (1 - \phi)D_i}, \quad \check{\mathbf{v}} = \phi \check{\mathbf{v}}_r + \check{\mathbf{v}}_s.$$

- 7: Explicitly marching forward the transport system (??) to next time $n + 1$ for $\mathcal{U}^{n+1}$.
 - 8: **end for**
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1.1.1 Discontinuous Porosity

Due to the presence of no melt regions and different phase regions, we may obtain different reconstructed values of dimensionless enthalpy and mass concentration from two sides of an edge shared by two elements and thus two different calculated porosities. However, it is crucial to have one compatible and appropriate porosity value used on the element edge in the conforming MFEM scheme.

Therefore, we evaluate the geometric average of the two values of porosity ϕ_e^+ and ϕ_e^- at each quadrature points on both sides of an element edge e as

$$\bar{\phi}_e = \frac{2\phi_e^+\phi_e^-}{\phi_e^+ + \phi_e^-}, \tag{1.1}$$

where ϕ_e^+ and ϕ_e^- are porosity evaluated from two sides of the element edge e . The geometric averaged porosity $\bar{\phi}_e$ is required for the continuous compaction equation in Darcy equation (??) , where face integral is replaced with edge integral through divergence theorem. When the porosity on one side equal zero, the geometric average

of the porosity equals zero as well, which highlights the fact that there is no liquid flow should travel through this edge.

1.1.2 Compaction Length

The concept of compaction length was first proposed by ?. The phenomenon of compaction is due to the pressure difference between phases, which is commonly parametrized by bulk viscosity $\zeta \sim \frac{\eta}{\phi}$.

1.1.3 Some Simplifications

We solve a simplified equation for the following sections.

- lithostatic pressure.
- Ommiting diffusion term for now.

1.2 Pseudo One-Dimensional Column Tests

In this section, we first test our coupled algorithm on a 2D $M \times N$ grid with a pseudo-1D column problem. We test two different scenarios, one couples MFEM and FVM solver without phase computation but with a simple prescribed melting function $\Gamma(\mathbf{x})$ instead. The other one exhibits a fully coupled scenario where phase transition and porosity evolution present. For the two tests presented below, we fix mesh resolution as $M = 2$ and $N = 100$. We fix the parameter $\Theta = 0$ as well.

1.2.1 Prescribed Melting Case

Before coupling mechanical, transport and phase package altogether into one full computation, we illustrate a simpler model coupling MFEM and FVM solver without involving phase package. Instead of using phase package computation to decide porosity distribution, we prescribe the melting rate or the change of porosity

as a fixed source function distributed in space as

$$\phi(t) = \Gamma(\mathbf{x}). \quad (1.2)$$

We write conservation of liquid mass with the prescribed melting term Γ (1.2) as

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\phi \mathbf{v}_l) = \Gamma, \quad (1.3)$$

where the diffusion term is being dropped for simplicity in this case.

We prescribe the following boundary conditions for MFEM and FVM solver respectively. For the Darcy-Stokes MFEM solver:

- Bottom Edge $y = -H$;
 - Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v, \quad |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e v \, ds.$$

- Tangential Stokes velocity is prescribed with essential boundary condition:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$

- Normal Darcy flux is prescribed with essential boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} \, ds = |e|^{-1} \int_e u \, ds = 0.$$

- Top Edge $y = 0$;
 - Normal Stokes velocity and supplemental edge averaged flux are prescribed with natural boundary conditions: traction free

$$t_0 = 0;$$

- Tangential Stokes velocity is prescribed with zero essential boundary condition:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\tau} = 0.$$

- Normal Darcy flux is prescribed with natural boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e u ds = 0.$$

- Left and Right Edges $x = \pm L$;

- Normal Stokes velocity and supplemental edge averaged flux are prescribed with essential boundary conditions:

$$\check{\mathbf{v}}_s \cdot \boldsymbol{\nu} = v = 0, |e|^{-1} \int_e \mathbf{b} \cdot \boldsymbol{\nu} ds = 0;$$

- Tangential Stokes velocity is prescribed with the natural boundary condition: free traction

$$t_0 = 0;$$

- Normal Darcy flux is prescribed with non-permeable essential boundary condition:

$$|e|^{-1} \int_e \phi \check{\mathbf{v}}_r \cdot \boldsymbol{\nu} ds = |e|^{-1} \int_e u ds = 0.$$

For FVM solver:

- Bottom Edge $y = -H$: A Dirichlet boundary condition is prescribed as inflow;
- Top Edge $y = 0$: A free outflow boundary condition is prescribed;
- Left and Right Edges $x = \pm L$: A no flow boundary condition is prescribed.

For simplicity and efficiency in implementation and implementation and computation, we use one-dimensional ML-WENO(3,2) stencils as depicted in Figure 1.1, since we don't need excessive resolution and order of accuracy higher than one in x direction. We use second order SSP Runge-Kutta time integration scheme. We show two different scenarios equipped with different melting functions and initial conditions:

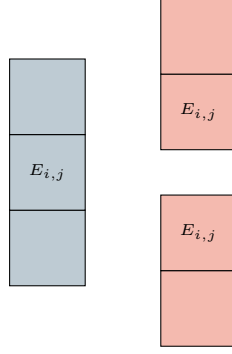


Figure 1.1: An exhibition of a set of 1D ML-WENO (3,2) stencils on a rectangular mesh.

Case 1: A piece-wise constant source function.

$$\Gamma_1(z) = \begin{cases} 2.5e - 3, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \quad (1.4)$$

We assign a zero initial condition here as

$$\phi_1(z, 0) = 0.0, \quad z \in [-2.0, 0.0], \quad (1.5)$$

Figure 1.2 illustrates the evolution of porosity, Stokes and Darcy velocities, Stokes and Darcy pressure, as well as the effective pressure, defined as $p_{\text{eff}} = p_s - p_l$. The third figure, corresponding to time 1325, demonstrates that the system has reached a steady state after a period of evolution.

Case 2: We examine a slightly more complicated case where a melting source trapped under the bottom of the upwelling mantle column as

$$\Gamma_2(z) = \begin{cases} 0, & -1.5 < z \leq 0, \\ 1.5e - 3, & -1.8 < z \leq -1.5, \\ 0, & -2.0 \leq z \leq -1.8, \end{cases} \quad (1.6)$$

where we assign an initial condition with a porosity channel as

$$\phi_2(z, 0) = \begin{cases} 5e - 2, & -1.5 < z \leq 0, \\ 0, & -2.0 \leq z \leq -1.5. \end{cases} \quad (1.7)$$

1.2.2 Phase Transition Case

1.3 Two-Dimensional Test

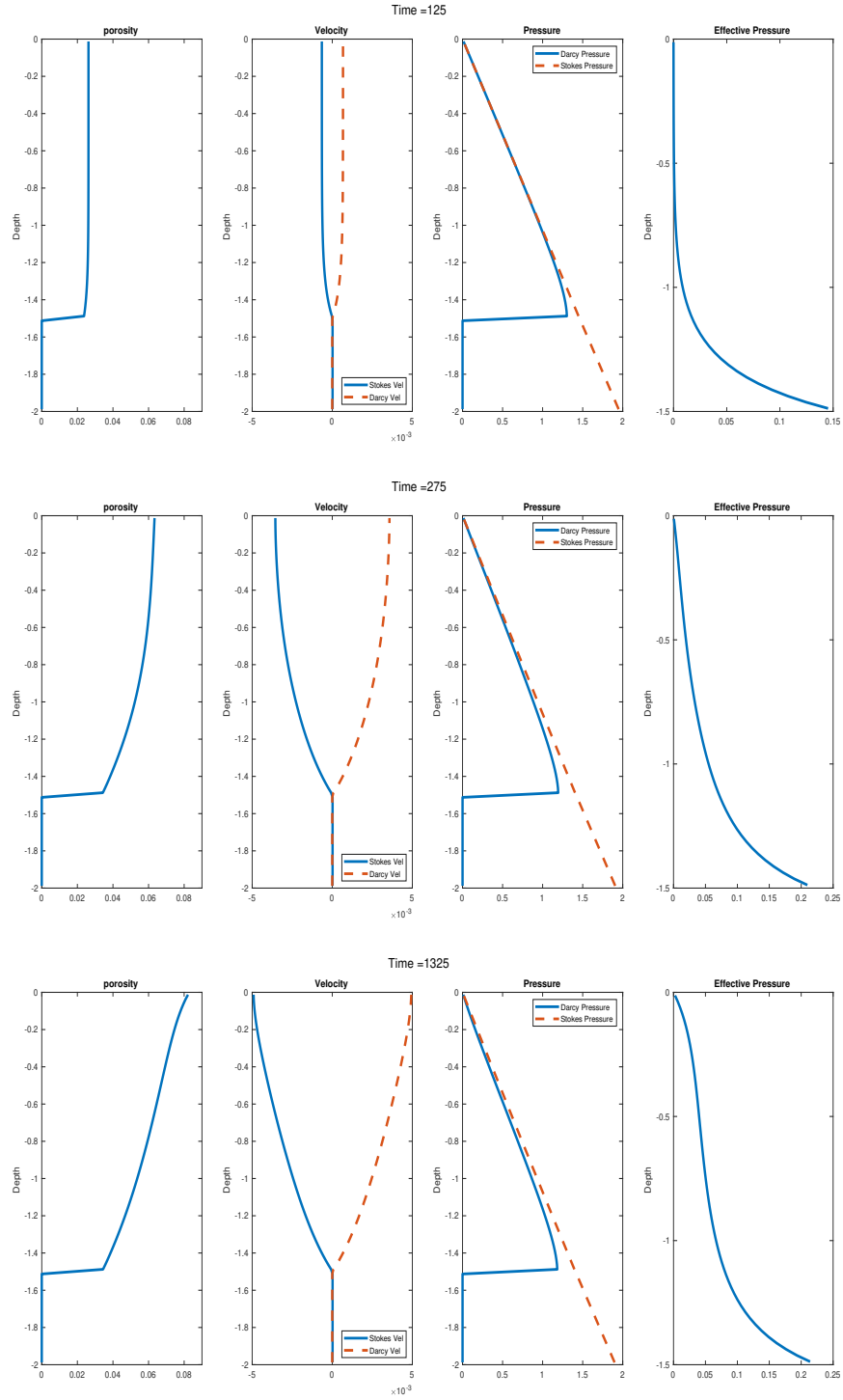


Figure 1.2: Prescribed melting case with constant melting source term 1.4. Shown at time = 125, 275 and 1325, where the system reaches steady state at the time = 1325.

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